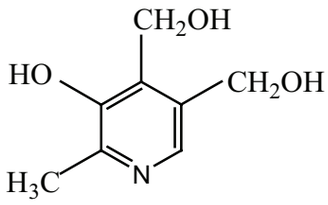
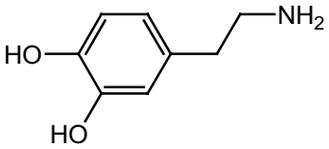
INTRODUCTION

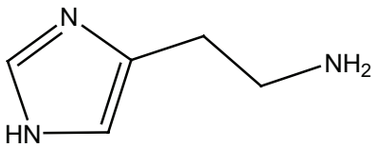
Organic compounds containing nitrogen are essential to life. For example; amines, the organic derivatives of ammonia play an important role in bioregulation, neurotransmission,etc., Pyridoxine, Vitamin B_6 is an organic nitrogen compound which is needed to maintain the health of nerves, skin and red blood cells. Plants synthesise alkaloids, and biologically active amines to protect them from being eaten away by insects and other animals. Diazonium salts finds important applications in synthetic organic chemistry. Nitrogen compounds are the important constituents of explosives, drugs, dyes, fuels, polymers, synthetic rubbers, etc.,



Vitamin B_6



dopamine
neurotransmitter



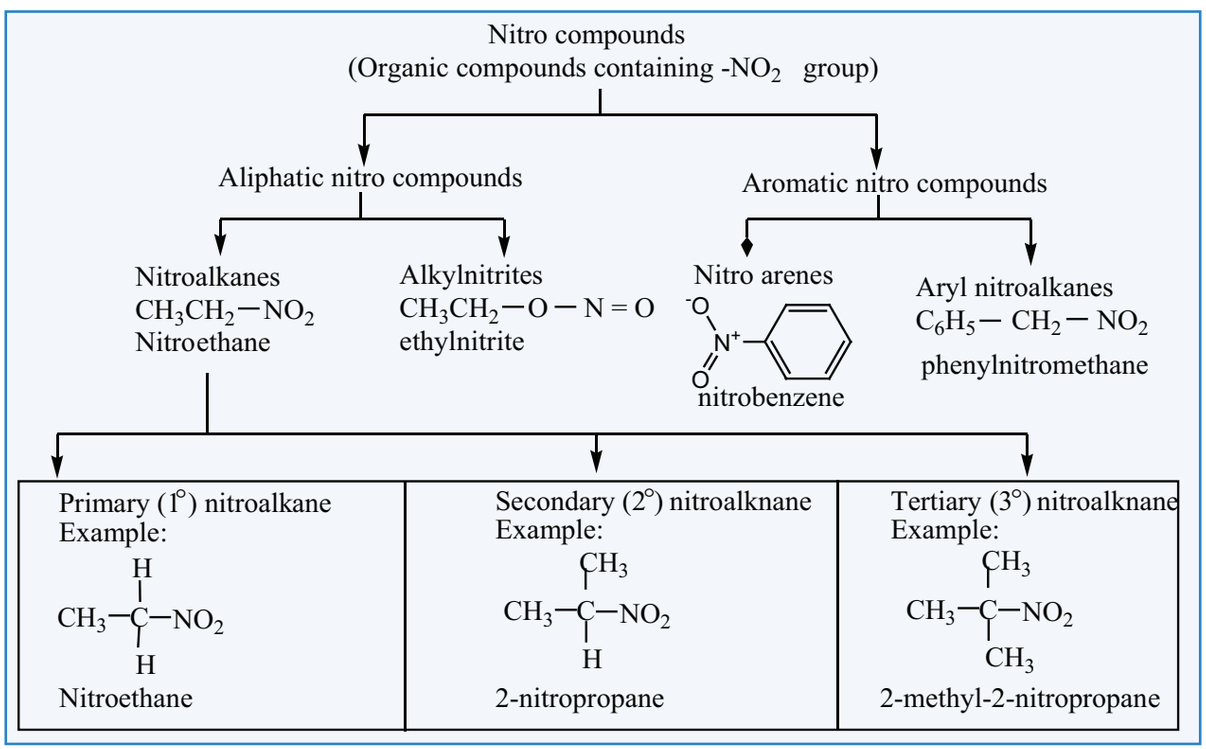
Histamine
dilates blood vessels

In this unit, we will learn the preparation, properties and uses of nitrocompounds and amines.

13.1 NITRO COMPOUNDS

Nitro compounds are considered as the derivaties of hydrocarbons. If one of the hydrogen atom of hydrocarbon is replaced by the $-NO_2$ group, the resultant organic compound is called a nitrocompound.

13.1.1 Classification of nitrocompounds



XII U13-Organic Nitrogen Compounds.indd 197 16-12-2022 19:45:03